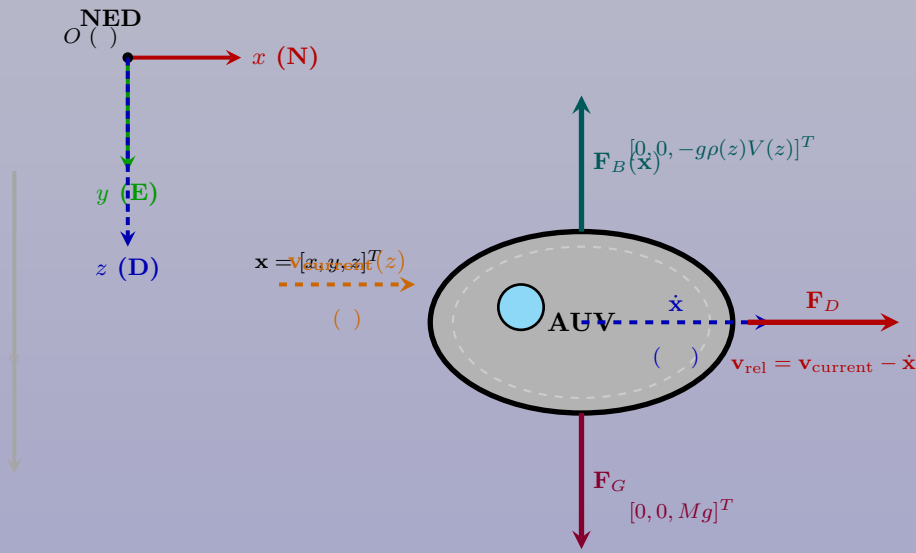




$\mathbf{M}_{sys}:$

$\mathbf{M}_{sys} =$

$$\begin{bmatrix} M + m_{a,xy} & 0 & 0 \\ 0 & M + m_{a,xy} & 0 \\ 0 & 0 & M + m_{a,z} \end{bmatrix}$$

- $(\neq)$
- $\rho(z):$
- $\rho \approx 1000 + 0.8(S - 35) - 0.2(T - 20) + 0.004z$
- $S \quad T \quad z$
- $V(z):$
- $V(z) = V_0(1 - \gamma z)$
- γ ,
- $\mathbf{F}_D:$
- $\mathbf{F}_D = \frac{1}{2}\rho(z)\mathbf{C}_D \odot \mathbf{A} \odot (|\mathbf{v}_{rel}| \circ \mathbf{v}_{rel})$
- ,

$$\mathbf{M}_{sys} \cdot \ddot{\mathbf{x}} = \mathbf{F}_G + \mathbf{F}_B(\mathbf{x}) + \mathbf{F}_D(\mathbf{x}, \dot{\mathbf{x}})$$

- :
- $\mathbf{F}_G:$ ()
 - $\mathbf{F}_B:$ ()
 - $\mathbf{F}_D:$ ()

2025 8 1-7